

30-Day Python GitHub Project Challenge

A beginner-to-intermediate roadmap for building a professional Python GitHub portfolio

Goal: Complete practical Python projects every day, document them properly, and build consistent GitHub activity.

Recommended approach: Spend 1–2 hours per project. Focus on understanding the code, testing it, writing a README, and making a meaningful Git commit.

30-Day Project Roadmap

Day	Project	Main Concepts
1	Student Profile Generator	Variables, data types, input/output
2	Simple Bill Calculator	Arithmetic, input, formatting
3	Temperature Converter	Functions, conditions
4	Age & Personal Information Calculator	Datetime, calculations
5	Even/Odd & Number Analyzer	if/else, operators
6	Simple Calculator	Functions, operators
7	Number Guessing Game	Loops, random
8	Rock Paper Scissors	Conditions, random
9	Password Generator	Strings, random
10	Password Strength Checker	Conditions, strings
11	To-Do List CLI	Lists, functions
12	Contact Book	Dictionaries, CRUD
13	Quiz Game	Lists, dictionaries, scoring
14	Expense Tracker	Lists, dictionaries, file handling
15	Student Grade Management System	Functions, lists, dictionaries
16	Bank Account Simulator	OOP, classes
17	Library Management System	OOP, CRUD
18	Inventory Management System	OOP, file handling
19	File Organizer	os, shutil, automation
20	Bulk File Renamer	OS automation
21	CSV Data Analyzer	CSV + Pandas
22	Sales Data Analyzer	Pandas
23	Student Performance Analyzer	Pandas + Matplotlib
24	Weather Data Analyzer	APIs + JSON
25	GitHub Profile Analyzer	API + JSON
26	Web Scraper	Requests + BeautifulSoup
27	Personal Finance Dashboard	Pandas + Matplotlib
28	Attendance Management System	Python + CSV
29	Job Application Tracker	Python + SQLite
30	Student Productivity Dashboard	Python + Pandas + SQLite + Charts

GitHub Repository Structure

30-days-python-projects/

■ ■ ■ ■ Day-01-student-profile-generator/

■ ■ ■ ■ main.py

■ ■ ■ ■ README.md

■ ■ ■ ■ Day-02-bill-calculator/

■ ■ ■ ■ main.py

■ ■ ■ ■ README.md

■ ■ ■ ■ ...

■ ■ ■ ■ Day-30-productivity-dashboard/

■ ■ ■ ■ app.py

■ ■ ■ ■ data/

■ ■ ■ ■ requirements.txt

■ ■ ■ ■ README.md

■ ■ ■ ■ README.md

Daily GitHub Checklist

■ Create the project folder

■ Write and test the Python code

■ Add comments/docstrings where useful

■ Create a README.md explaining the project

■ Add screenshots or sample output when useful

■ Add requirements.txt when external libraries are used

■ Commit the work with a clear message

■ Push the project to GitHub

Suggested Commit Messages

Day 01: Add Student Profile Generator

Day 02: Add Bill Calculator

Day 03: Add Temperature Converter

...

Day 30: Add Student Productivity Dashboard

Skill Progression

Period	Focus	Skills
Days 1–7	Python fundamentals	Variables, input/output, conditions, loops, functions
Days 8–14	Logic + CLI applications	Random, strings, lists, dictionaries, file handling
Days 15–20	OOP + automation	Classes, CRUD, os, shutil, automation
Days 21–27	Data + APIs	Pandas, Matplotlib, APIs, JSON, web scraping
Days 28–30	Real-world applications	CSV, SQLite, dashboards, data visualization

Final target: Finish 30 projects with clean code, documentation, meaningful commits, and a GitHub repository that demonstrates progression from Python fundamentals to practical data-driven applications.